

Question D.2 — Why Kendall’s τ_b and Not Pearson or Spearman

This section justifies the association coefficient used throughout Q D. The counter-example of Section 4 comes from `scripts/contre_exemple_kendall.py`, which computes both coefficients twice, once from their definition and once through `scipy`, so that each controls the other. Census figures come from the campaign of 2000 runs (20 amplitudes $\times$ 100 seeds, $H = 2000$, $M = 10$ trees in the adaptive random forest, ARF); the bootstrap confidence intervals (CI) of Section 1 are written by `scripts/derives_QD.py`.

1 Three Features of the Data

The quantities to be correlated are durations measured on a finite horizon $H = 2000$. Three features bear on the choice of coefficient.

Non-monotonicity. The median τ_{ARF} is 118 steps at $\Delta e = 0.028$, rises to 346 at $\Delta e = 0.085$, then falls steadily to 28 at the top of the grid: a range of 28 to 346, a factor of 12.4, and not monotone in Δe . The rise is not an artefact of the median. The means move the same way (162.7 then 385.9), and the bootstrap confidence intervals of the two medians are disjoint, [92.5; 158.5] against [281; 455] (Table 1). The *location* of the peak is less certain: the interval at $\Delta e = 0.141$, [231.5; 307], overlaps the one at $\Delta e = 0.085$ over [281; 307], so the two cells are not separated by the rule used here and the claim is that a peak exists at low amplitude, not that it sits at one particular cell. That negative statement is weak, and the control quantifies how weak: at a ratio of medians of 1.32, which is exactly 346/263, the rule separates two samples of 100 in only 23.4% of 500 repetitions on a log-normal law. Failing to separate these two cells is therefore close to uninformative, whereas the separation of $\Delta e = 0.028$ from $\Delta e = 0.085$ is not: the rule falsely separates two samples of the same law in 1.0% of repetitions. The relationship is therefore worse than merely curved, which would already understate Pearson’s r . No single linear coefficient summarises it, and its sign depends on which part of the grid is included. This matches what is established elsewhere about the power-law fit of the article, whose measured-to-predicted ratio is 0.19 at the lowest amplitude and 2.08 at $\Delta e = 0.141$: the fit holds only mid-grid. Figure 1 shows the full distribution at every amplitude.

Δe	median τ_{ARF} (steps)	95% CI	mean	seeds
0.028	118	[92.5; 158.5]	162.7	100
0.085	346	[281; 455]	385.9	100
0.141	263	[231.5; 307]	331.9	100
0.498	28	[27; 29]	29.4	100

Table 1: Median of τ_{ARF} with its percentile bootstrap confidence interval (CI), 10 000 resamples of the 100 seeds of the amplitude, generator seed 0. The row at $\Delta e = 0.141$ is there so that the overlap discussed above can be checked. Table `QCD_ic_medianes_tau_arf`. Before being applied to the data the estimator passed five controls on cases with a known answer, stored in `QCD_ic_medianes_temoins`: on 500 samples of size 100 from a log-normal law of known median, the interval covered the true median in 95.2% of cases; on 500 pairs of samples from the same law, the interval of the difference of medians contained 0 in 96.4% of cases; on a constant sample the interval collapsed to the point; and the two controls that bear on the rule actually used here, namely declaring two medians distinct when their intervals are disjoint, gave a false-separation rate of 1.0% under the null and a separation rate of 23.4% at a ratio of medians of 1.32.

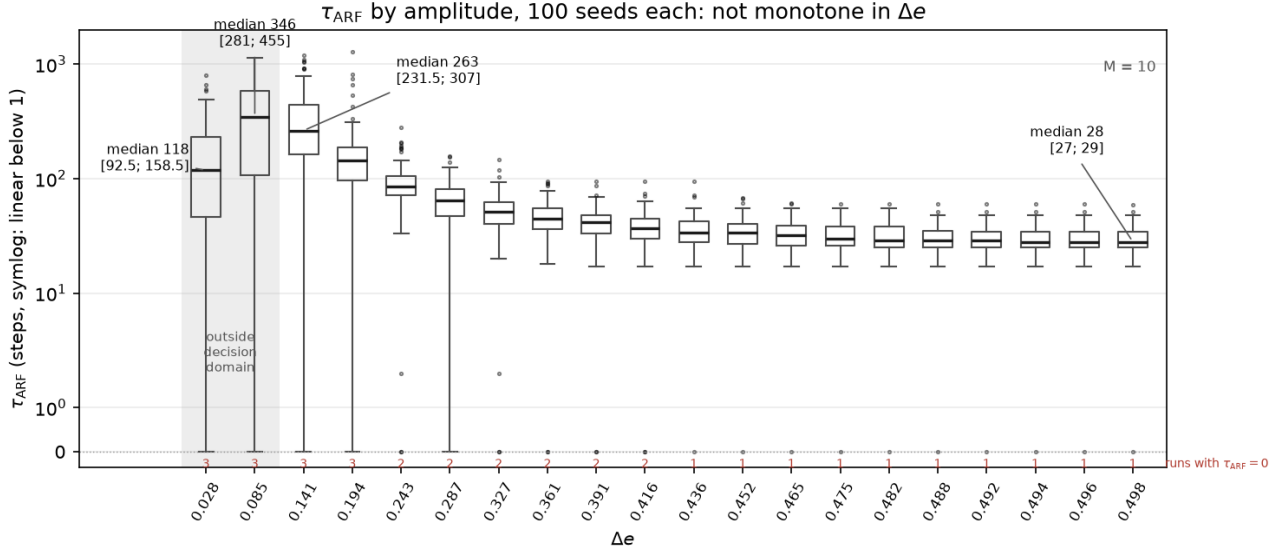


Figure 1: τ_{ARF} on the 100 seeds of each amplitude; the four medians of Table 1 are annotated with their interval. The axis is symmetric logarithmic, linear below 1: a plain logarithmic axis would silently drop the 34 runs in which a tree is replaced at the change point itself ($\tau_{\text{ARF}} = 0$), and the box would rest on 97 to 99 seeds while the title claims 100. Their count per amplitude is written in red on the zero line; they occur at all 20 amplitudes, three at the lowest and one at the highest. What to see: the median rises from the first amplitude to the second before it falls, the spread shrinks by more than an order of magnitude across the grid, and the two shaded amplitudes lie outside the decision domain of Q D.4.

Extreme values. At low amplitude the durations are large and widely dispersed. Pearson’s coefficient is a ratio of moments and is not robust, so a handful of runs in the tail move it substantially. Both rank coefficients are unaffected.

Administrative censoring at H . A duration never reached within the horizon has no numerical value. $\tau_{\text{swap}}(75\%)$ is censored in 9.10% of runs overall and in up to 78% at a single amplitude; τ_{det} at $\lambda = 50$ is censored in 99.95%.

Which survives stratification. The subject invites the reader to check what remains once Δe is held constant within a stratum. The first feature does not survive: the non-monotonicity is non-monotonicity *in* Δe , and within a stratum Δe is fixed by construction. The other two survive, and they are objections of different kinds. Extreme values make Pearson’s coefficient inefficient: it can still be computed, it is merely unstable and moves with a few runs. Censoring makes it undefined, because a censored run has no numerical value to enter the sum of products, so the coefficient cannot be formed without first inventing one. The second objection carries the argument, since the first is a matter of degree and the second a matter of definition. It is also the objection that bites on our data, where $\tau_{\text{swap}}(75\%)$ is censored in nearly four runs out of five at $\Delta e = 0.085$.

Both rank coefficients handle censoring the same way and for the same reason. The horizon H is identical for every run, so a censored duration genuinely exceeds every observed one and censored runs may be placed at the last rank, tied among themselves. The argument would collapse under a variable horizon, where a censored value is only known to exceed *its own* horizon and no ranking follows. That is where the choice between Spearman and Kendall becomes decisive.

2 Definition of τ_b

For n pairs, let C be the number of concordant pairs, D the discordant ones, and $n_0 = n(n-1)/2$. Let t_x count the pairs tied on x only and t_y those tied on y only, pairs tied on both sides being counted in each. Then

$$\tau_b = \frac{C - D}{\sqrt{(n_0 - t_x)(n_0 - t_y)}}. \quad (1)$$

The denominator distinguishes τ_b from the ordinary τ_a : each tie block is removed from the count on the side where it occurs, so the coefficient is normalised by the pairs that could in principle have been ordered on both sides. Spearman’s ρ has no such correction. It is Pearson’s coefficient computed on mid-ranks, and a tie block simply receives a common mid-rank that then enters a sum of squares as an ordinary value.

3 Two Tie Structures, Treated Separately

The comparators of Q D do not present the same tie structure, and the consequence differs.

Ties on both sides: τ_{ARF} against $\tau_{\text{swap}}(q)$. Both are durations and both can be censored at H . Censored runs form a tie block on the y side. τ_{ARF} is never censored (0.0000), but it is discrete and takes only 321 distinct values over 2000 runs, so it carries tie blocks of its own. Equation (1) corrects on both sides and the compensation is complete: the coefficient is normalised by exactly the set of orderable pairs.

Ties on one side: τ_{ARF} against $A(H)$ and $S_{\text{max}}(H)$. These two are defined on every run and never censored: 2000 values out of 2000, with 1389 and 1544 distinct values respectively. There is no tie block on their side at all, so the correction applies to the x side only. The denominator still counts the orderable pairs, but the two sides are no longer symmetric: ties in τ_{ARF} remove pairs from the numerator without any counterpart, and the coefficient is displaced *towards zero* relative to a tie-free measurement of the same association. That direction cuts both ways for Q D.3: a discernible coefficient is if anything understated, but a coefficient indistinguishable from zero could in principle be a real association pulled towards zero by the ties. On this campaign the effect is negligible either way: within a stratum of the decision domain only 0.2% to 4.5% of the pairs are tied on τ_{ARF} , which caps $|\tau_b|$ at 0.977 or more.

4 The Counter-Example, and an Error in the Prescribed Exercise

Ten fictitious runs on a common horizon $H = 2000$: four observed, six censored. The exercise as prescribed asks for the four observed values to be perfectly ordered. Carried out literally, with the censored runs tied on both sides, it yields $\rho = \tau_b = 1.000000$ exactly. The gap is zero and the counter-example demonstrates nothing.

The reason is sharper than “the example is badly chosen”. Ties on both sides require τ_{ARF} itself to be censored, and on this campaign it never is: `censored_arf` = 0.0000 on all 2000 runs. The configuration under which the prescribed exercise produces no gap cannot occur in the data it is meant to illustrate.

Configuration	ρ (Spearman)	τ_b (Kendall)	gap $\rho - \tau_b$
Prescribed, read literally (ties both sides)	1.000000	1.000000	0.000000
Observed values ordered, τ_{ARF} uncensored	0.887625	0.816497	0.071129
Observed values anti-ordered	0.751068	0.489898	0.261170

Table 2: Ten runs, four observed and six censored at H . Each coefficient is computed from its definition and by `scipy`; the two agree to 2.2×10^{-16} or better in every cell, which is the control on the calculation.

Two things separate the three rows, and they should not be conflated. What removes the gap entirely is the tie structure: as soon as τ_{ARF} is left uncensored, which is the situation that actually holds, the two coefficients part company by 0.071 even with the observed runs perfectly ordered. What then amplifies the gap is the ordering of the observed runs: with four anti-ordered observations, Spearman reports 0.751 where τ_b reports 0.490, a gap of 0.261, nearly four times the ordered case.

The mechanism is the same in both rows: the six censored runs share a mid-rank that enters Spearman’s sums of squares as an ordinary value, while τ_b removes those pairs from both counts. The size of the gap is not, by itself, evidence of an artefact: for a bivariate normal law with correlation r , $\tau = (2/\pi) \arcsin r$ and $\rho_S = (6/\pi) \arcsin(r/2)$, so ρ_S runs about 1.5 times τ for weak associations

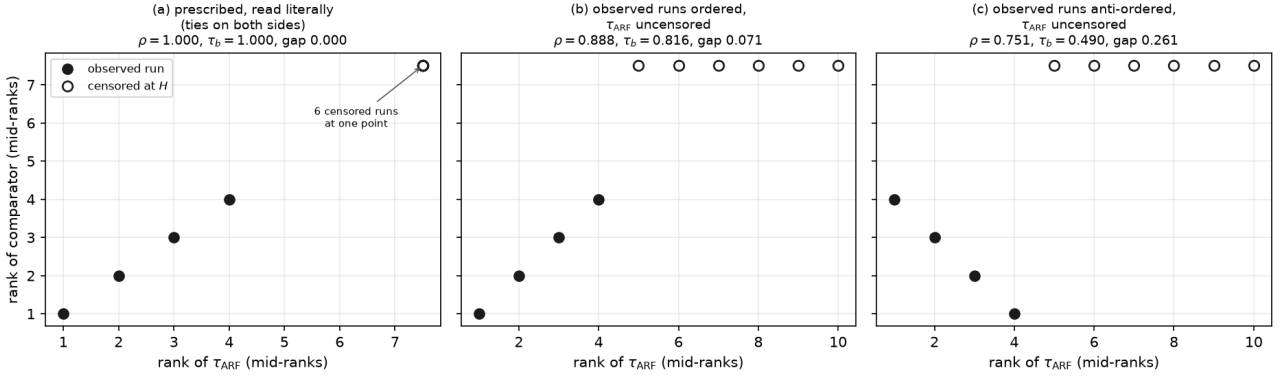


Figure 2: The three configurations of Table 2, as mid-ranks. Filled markers: observed runs; hollow markers: runs censored at H , which share one mid-rank on the comparator side. (a) The prescribed exercise read literally also ties the six censored runs on the τ_{ARF} side, and the two coefficients coincide. (b) With τ_{ARF} left uncensored, the same four ordered observations already separate Spearman from Kendall by 0.071. (c) Anti-ordering the four observations widens the gap to 0.261. Table QCD_contre_exemple_kendall.

even without any tie. The prescribed exercise hides this by silently assuming ties on both sides, which cancels the effect exactly.

5 Conclusion

Pearson is excluded by definition rather than by degree, since censored runs carry no numerical value. Spearman is computable, but its value has no direct reading and its handling of the tie blocks created by the censoring is implicit. Kendall's τ_b , which reads on pairs and corrects for ties explicitly, is used throughout Q D, with censored runs imputed to the last rank, which is legitimate because the horizon is common to all runs and only for that reason.